

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	Karthik K	Reg. No.:	192525040
Section / Batch:	RA	Date:	10/07/2026

Code of Conduct

I Karthik K / 192525040 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Karthik K

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

CO1_AT1 – C Code Output Prediction – Answer Sheet

Karthik K | Reg. No.: 192525040 | Section: RA

Q1 C Code – Array Traversal

10 marks

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Answer (Output): 50

$sum = 5+10+15+20 = 50$

Screenshot Proof:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser address bar shows the URL: simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "KONIJETI KARTHIK" with the registration number "192525040 RA". The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION" and features a "Back" button. On the left, a "QUESTIONS" sidebar lists five questions, each with a green checkmark and a "10 marks" label. The selected question, "Q1 C Code – Array Traversal", is displayed in the main area. The C code is shown in a text editor, and the "Your Answer" field contains the value "50".

QUESTIONS

- Q1 C Code – Array Traversal 10 marks
- Q2 C Code – Linear Search 10 marks
- Q3 C Code – Binary Search Mid 10 marks
- Q4 C Code – Pointer with Array 10 marks
- Q5 C Code – Array Update 10 marks

Q1 C Code – Array Traversal 10 marks

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Your Answer

50

Q2 C Code – Linear Search

10 marks

```
#include <stdio.h>
int main() {
int arr[] = {2,4,6,8,10};
int key = 8;
int i;
for(i=0;i<5;i++){
if(arr[i]==key){
printf("%d",i);
break;
}
}
}
```

Answer (Output): 3

8 is found at index 3 (0-based)

Screenshot Proof:

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- Q1: C Code – Array Traversal
- Q2: C Code – Linear Search (selected)
- Q3: C Code – Binary Search Mid
- Q4: C Code – Pointer with Array
- Q5: C Code – Array Update

The main area displays the selected question, "Q2 C Code – Linear Search", with the following C code:

```
#include <stdio.h>
int main() {
int arr[] = {2,4,6,8,10};
int key = 8;
int i;
for(i=0;i<5;i++){
if(arr[i]==key){
printf("%d",i);
break;
}
}
}
```

Below the code, the "Your Answer" field contains the value "3".

Q3 C Code – Binary Search Mid

10 marks

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Answer (Output): 3

$mid = (0+7)/2 = 3$ (integer division)

Screenshot Proof:

The screenshot shows a web browser window with the URL `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab" with "DS PROGRAMMING" below it, and a user profile for "KONIJETI KARTHIK" with ID "192525040 RA". The main heading is "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, a "QUESTIONS" sidebar lists five items, each with a green checkmark: Q1 (C Code – Array Traversal, 10 marks), Q2 (C Code – Linear Search, 10 marks), Q3 (C Code – Binary Search Mid, 10 marks), Q4 (C Code – Pointer with Array, 10 marks), and Q5 (C Code – Array Update, 10 marks). The main content area displays "Q3 C Code – Binary Search Mid" with a 10 marks value. It contains the same C code as shown in the first block. Below the code, under the heading "Your Answer", the number "3" is entered in a text box.

Q4 C Code – Pointer with Array

10 marks

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Answer (Output): 30

**(p+2) accesses arr[2] = 30*

Screenshot Proof:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and the user's name "KONIJETI KARTHIK" with a dropdown arrow. The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION" and features a "Back" button. On the left, a "QUESTIONS" sidebar lists five questions, each with a green checkmark and "10 marks":

- Q1 C Code – Array Traversal
- Q2 C Code – Linear Search
- Q3 C Code – Binary Search Mid
- Q4 C Code – Pointer with Array (highlighted)
- Q5 C Code – Array Update

The main area displays the question "Q4 C Code – Pointer with Array" with the following code:

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Below the code, the "Your Answer" section shows the input "30".

Q5 C Code – Array Update

10 marks

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

Answer (Output): 4

$arr[1] = arr[0] + arr[2] = 1 + 3 = 4$

Screenshot Proof:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and the user's name "KONIJETI KARTHIK" with a dropdown arrow. The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION" and features a "Back" button. On the left, a "QUESTIONS" sidebar lists five questions, each with a green checkmark and a "10 marks" label. The selected question, "Q5 C Code – Array Update", is displayed in the main area. The code for the question is:

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

 Below the code, the "Your Answer" section shows the input "4".

Q6 C Code - Linked List Node**10 marks**

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Answer (Output): 20

n1.next points to n2, so n1.next->data = n2.data = 20

Screenshot Proof:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab" and "DS PROGRAMMING", and a user profile for "KONIJETI KARTHIK" with ID "192525040 RA".

On the left, a "QUESTIONS" sidebar lists six items, each with a green checkmark and "10 marks":

- Q1: C Code – Array Traversal
- Q2: C Code – Linear Search
- Q3: C Code – Binary Search Mid
- Q4: C Code – Pointer with Array
- Q5: C Code – Array Update
- Q6: C Code - Linked List Node

The main content area displays "Q6 C Code - Linked List Node" with a "10 marks" label. It contains the same C code as shown in the first block. Below the code, there is a "Your Answer" section with a text input field containing the value "20".

Q7 C Code - Time Complexity Loop

10 marks

```
for(i=0;i<n;i++)
```

```
printf("**");
```

The complexity is _____

Answer (Output): $O(n)$

Single loop running n times $\rightarrow$ linear time complexity

Screenshot Proof:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "KONIJETI KARTHIK" with a dropdown arrow. The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION" and features a "Back" button. On the left, a "QUESTIONS" sidebar lists five questions, each with a green checkmark and a "10 marks" label. The selected question, "Q7 C Code - Time Complexity Loop", is displayed in the main area. It contains the C code snippet:

```
for(i=0;i<n;i++)  
printf("**");
```

 followed by the text "The complexity is". Below the code, there is a "Your Answer" section with a text input field containing the letter "O". At the bottom of the question area, there are two buttons: "Save Answer" and "Next →".

Q8 C Code - Nested Loop

10 marks

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("**");
```

Answer (Output): $O(n^2)$

Two nested loops each running n times $\rightarrow$ quadratic time complexity

Screenshot Proof:

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "KONIJETI KARTHIK" with a dropdown arrow. Below the header, the page title is "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, there is a "QUESTIONS" sidebar with five items, each marked with a green checkmark: Q1 (C Code - Array Traversal, 10 marks), Q2 (C Code - Linear Search, 10 marks), Q3 (C Code - Binary Search Mid, 10 marks), Q4 (C Code - Pointer with Array, 10 marks), and Q5 (C Code - Array Update, 10 marks). The main content area displays the selected question, "Q8 C Code - Nested Loop", with a "10 marks" label. The question code is:

```
for(i=0;i<n;i++)  
for(j=0;j<n;j++)  
printf("**");
```

 Below the code, there is a "Your Answer" section with a text input field containing the letter "O". At the bottom of the question area, there are two buttons: "Save Answer" and "Next →".

Q9 C Code - Binary Search Complexity**10 marks**

```
while(low<=high){  
mid=(low+high)/2;  
}
```

Answer (Output): $O(\log n)$ *Search space halves each iteration $\rightarrow$ logarithmic time complexity***Screenshot Proof:**

The screenshot shows a web browser window with the SIMATS Online Lab interface. The browser's address bar shows the URL `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "KONIJETI KARTHIK" with a dropdown arrow. The main content area is titled "CO1_AT1_C CODE OUTPUT PREDICTION" and features a "Back" button. On the left, a "QUESTIONS" sidebar lists five questions, each with a green checkmark and a "10 marks" label. The selected question, "Q9 C Code - Binary Search Complexity", is displayed in the main area. It shows the C code snippet:

```
while(low<=high){  
mid=(low+high)/2;  
}
```

 Below the code, there is a "Your Answer" section with a text input field containing the letter "O". At the bottom of the question area, there are two buttons: "Save Answer" and "Next →".

Q10 C Code - Pointer Increment**10 marks**

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d", *p);
}
```

Answer (Output): 10*p++ moves pointer to a[1]; *p = 10***Screenshot Proof:**

The screenshot shows a web browser window with the URL `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=330`. The page header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab DS PROGRAMMING", and a user profile for KONIJETI KARTHIK. The main heading is "CO1_AT1_C CODE OUTPUT PREDICTION". On the left, a "QUESTIONS" sidebar lists five items, all marked with a green checkmark. The selected question, "Q10 C Code - Pointer Increment", is displayed in the main area with its code and a "10 marks" value. Below the code, the "Your Answer" section shows the input "10".

QUESTIONS

- Q1 C Code – Array Traversal 10 marks ✓
- Q2 C Code – Linear Search 10 marks ✓
- Q3 C Code – Binary Search Mid 10 marks ✓
- Q4 C Code – Pointer with Array 10 marks ✓
- Q5 C Code – Array Update 10 marks ✓

Q10 C Code - Pointer Increment 10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10